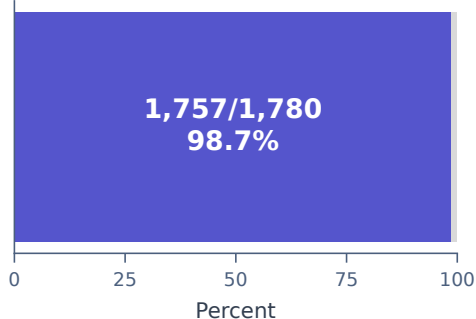
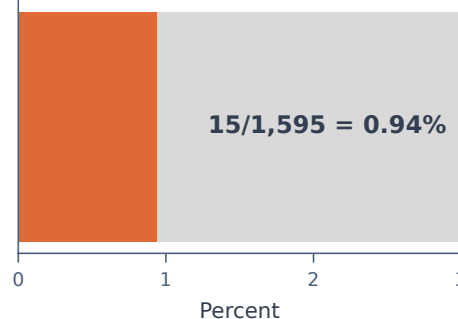


Technical validation of dominant repeat-length allele calling

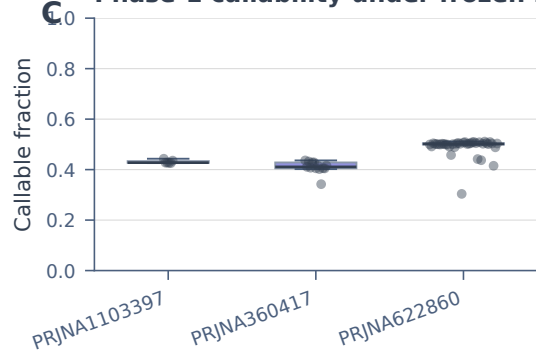
A Synthetic exact dominant allele



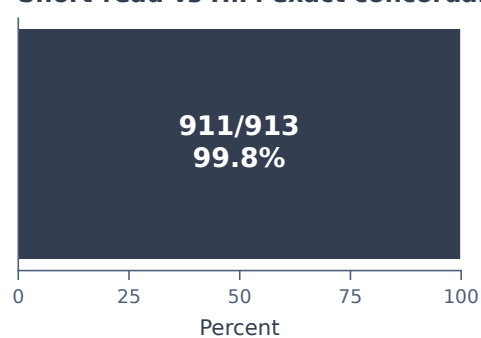
B False-confident wrong dominant allele



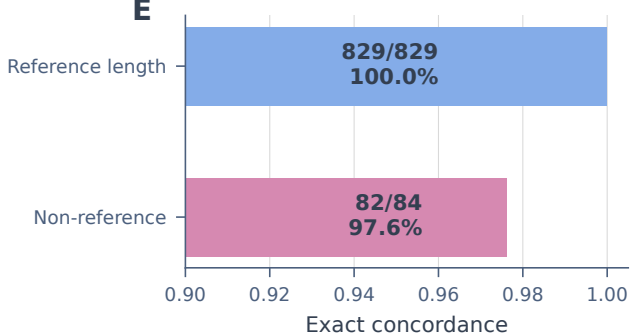
C Phase-1 callability under frozen rules



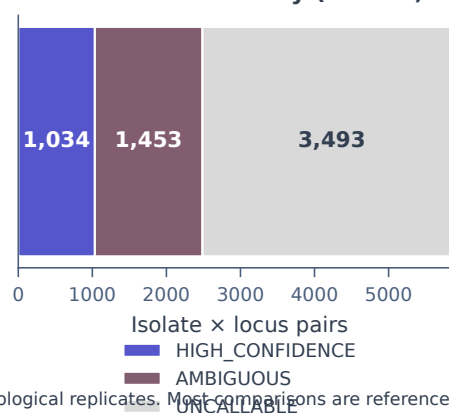
D Short-read vs HiFi exact concordance



E HiFi reference vs non-reference alleles



F HiFi truth availability (total 5,980)



HiFi values are descriptive per-comparison concordance, not 913 independent biological replicates. Most comparisons are reference-length; KNOWN_PV and mixture validation remain insufficient.